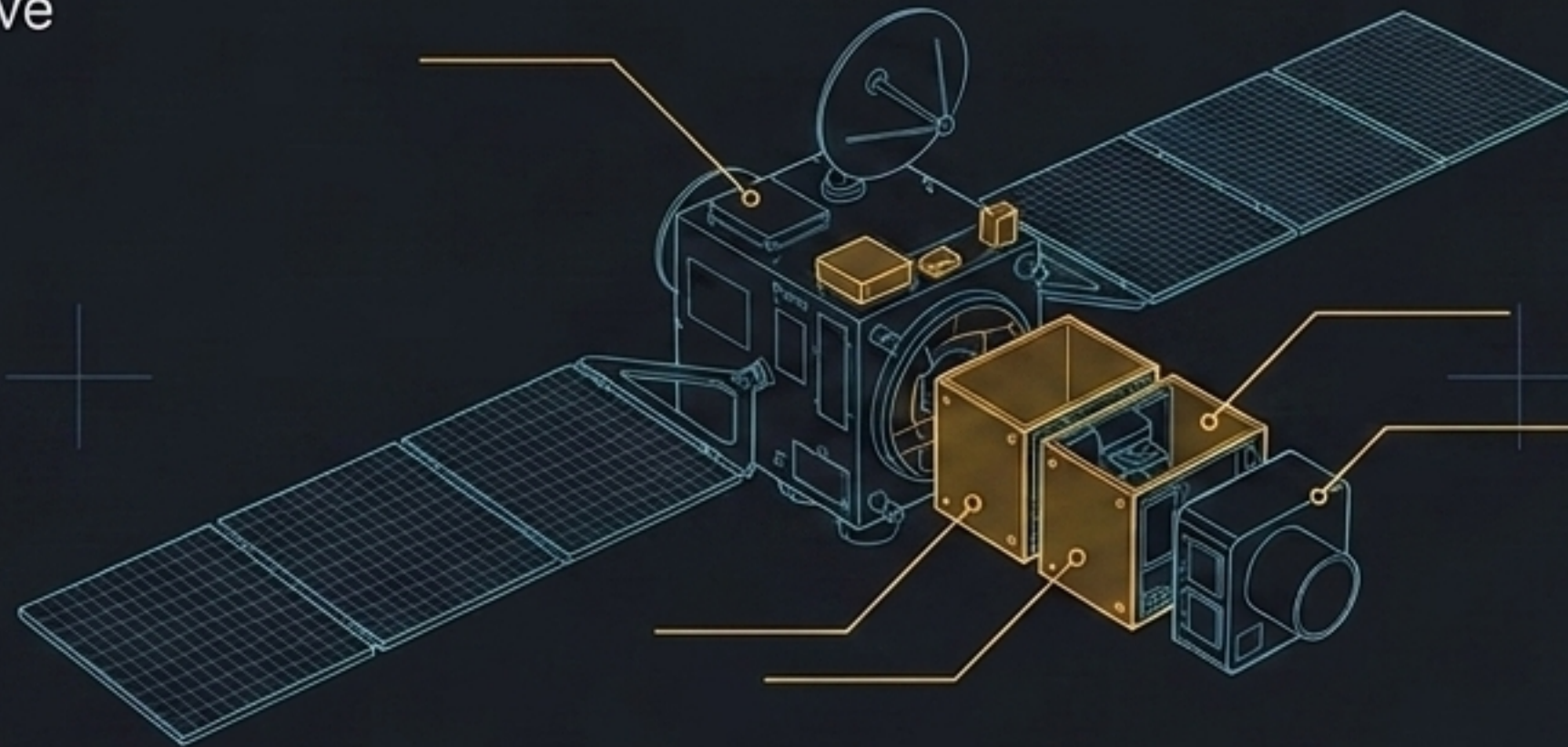


LFM-ORBIT: A Decentralized Operating System for Low Earth Orbit

A Decentralized Edge-Native Operating System for Low Earth Orbit



[PAYLOAD]

LFM2.5-VL-450M (450M production prototype • 1.6B planned upscale)

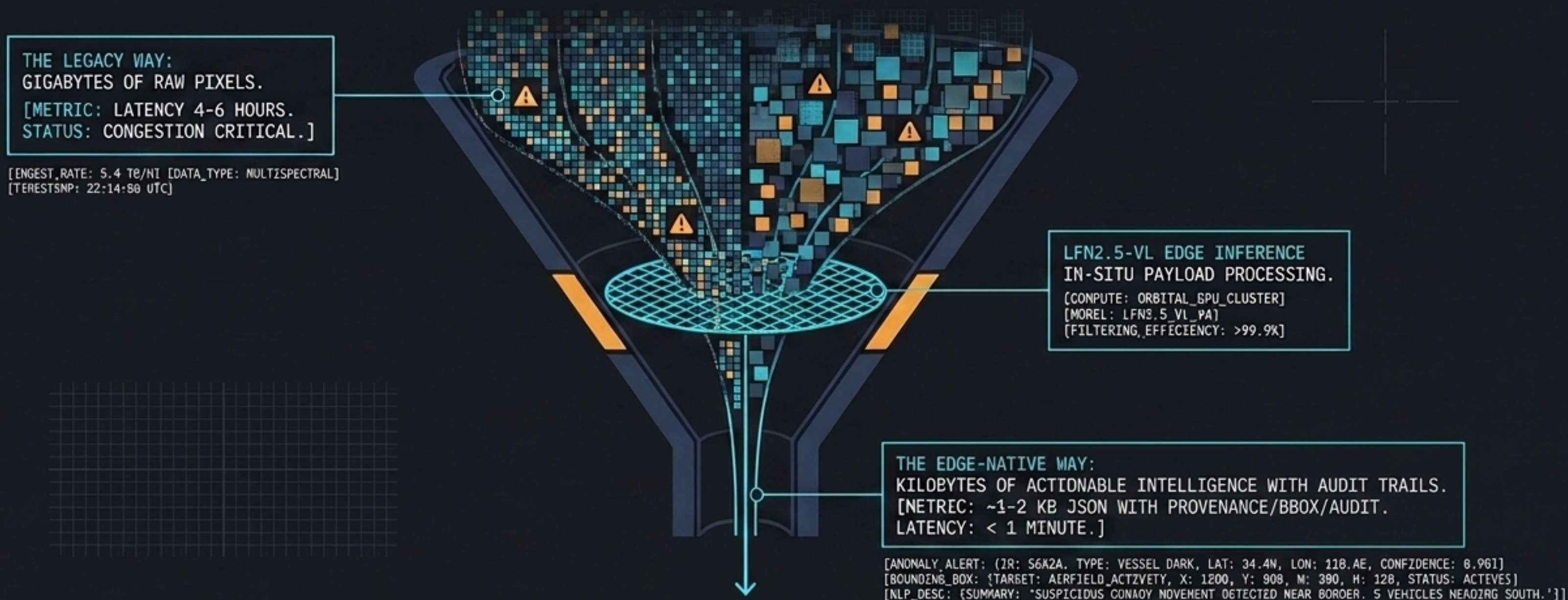
[ARCHITECTURE]

Dual-Agent Edge-Native

[TARGET]

Liquid AI x DPhi Space

The bent-pipe bottleneck cripples real-time response



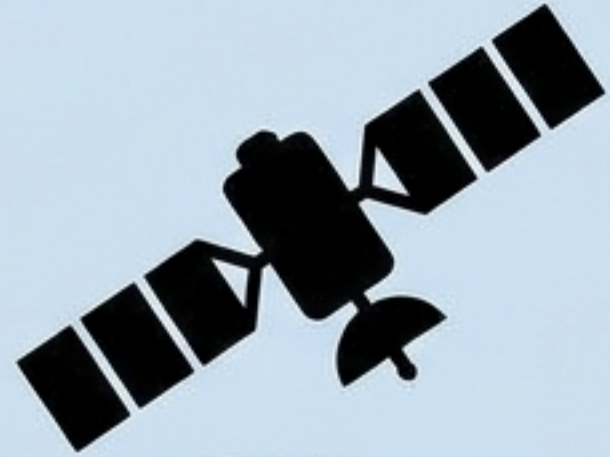
Satellites capture terabytes of data, but narrow downlinks create unacceptable latency for real-time emergencies. LFM-ORBIT solves this by moving the analyst into orbit, delivering ~1-2 KB structured JSON alerts with full provenance.

Edge-native infrastructure outcompetes centralized cloud pipelines

Dimension	Legacy Bent-Pipe		LFM-ORBIT
Processing Location	⚠	Terrestrial Ground Station	✓ In-Situ Payload / Edge
Downlink Payload	⚠	Gigabytes of raw .TIFF/video	✓ ~1–2 KB structured JSON alerts
Intelligence Latency	⚠	Hours to Days	✓ Seconds
Cloud Compute Cost	⚠	High recurring API/server fees	✓ Zero. Runs fully offline.
Data Privacy	⚠	Vulnerable third-party cloud routing	✓ Absolute. Browser-native local inference.

By shifting the computational burden to the hardware, LFM-ORBIT filters irrelevant environmental data before transmission.

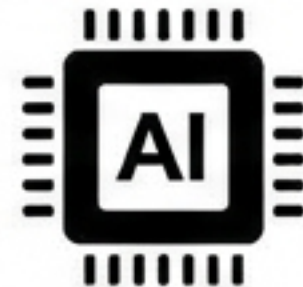
Orbital segment



DPhi Space
Clustergate Satellite



Target edge hardware
(Jetson Orin-class)
[COMPUTE_UNIT: NVIDIA_ORIN_NX]



LFM2.5-VL-450M (text-evidence
reasoning on pruned packets)
[MODEL_VER: LFM2.5-VL-450M]

Autonomous
Triage

Downlink

Ground Segment



Web Browser

willama WebAssembly engine & Local LLM

Open Source Basemaps

Zero-Backend
Client

Liquid Foundation Models unlock frontier intelligence on constrained edge hardware

MEMORY FOOTPRINT (LOWER IS BETTER)

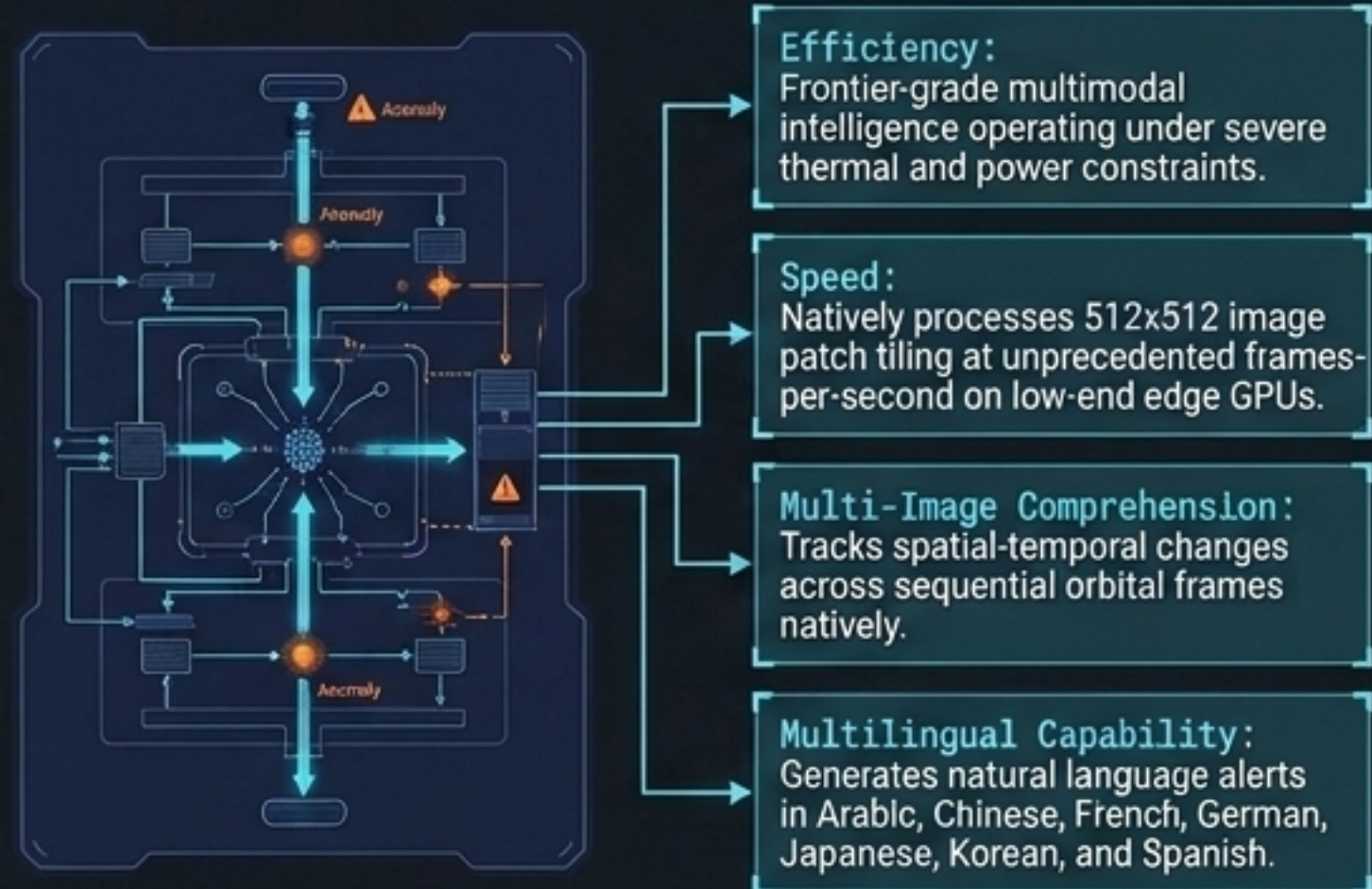
Massive headroom left on the 16GB Jetson Orin for OS, image buffering, and parallel processes.



FUTURE VARIANTS (PLANNED)

1.6B (Q4_0): ~2.2 GB
3B (Q4_0): ~3.0 GB
3B (Q8_0): ~4.2 GB

PERFORMANCE METRICS & ARCHITECTURE



NOTE: Performance metrics for 1.6B/3B future variants are subject to change. Current prototype uses 450M model.

DOSSIER

The shipped prototype uses a highly optimized 450M parameter variant. It is dynamically loaded for highly restrictive micro-satellite buses, consuming mere megabytes. Larger 1.6B and 3B models are planned future variants.

Perceptual deduplication culls 99.9% of redundant orbital telemetry

ANOMALY: QUEUED FOR DOWNLINK.

PRUNED:
1,483× REDUCTION IN PAYLOAD SIZE



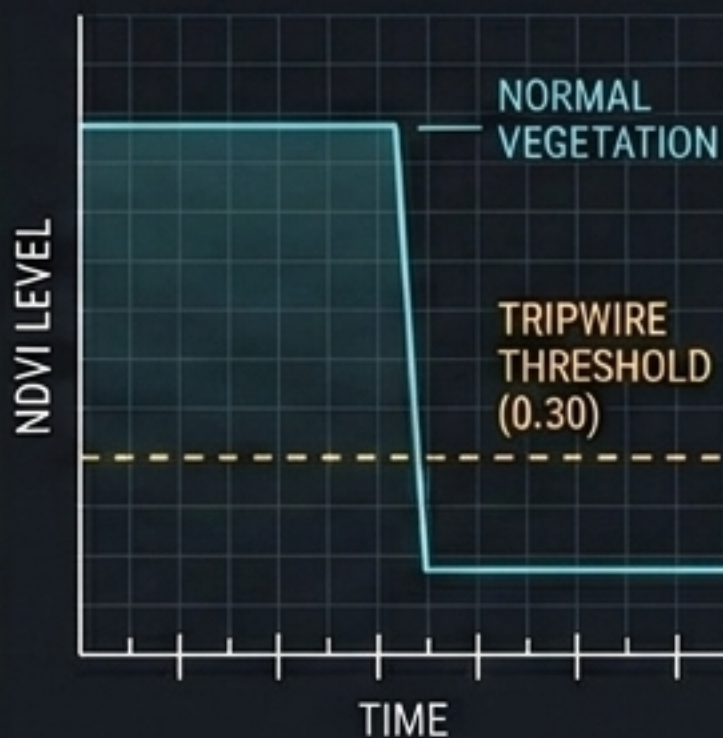
aHash: af3b9e7c

aHash: af3b9e7c

aHash: af3b9e7c

aHash: d9x2p11m

NDVI Tripwires



Only significant spectral deltas (NDVI drop > 0.30) wake the VLM from idle. The resulting ~1-2 KB alerts drastically conserve payload power.

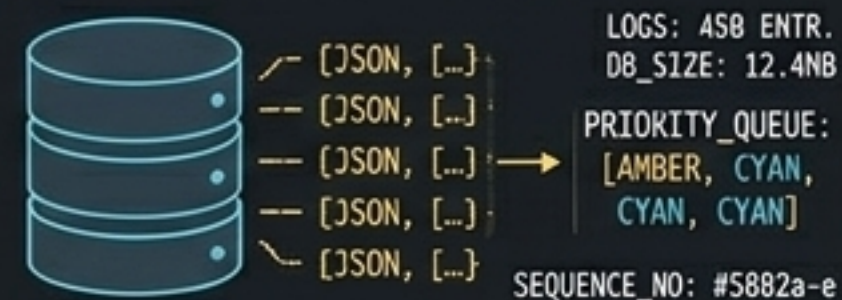
Store-and-forward queues guarantee delivery during communication blackouts

Step 1: Out-of-Contact Generation



Satellite detects an anomaly over the Pacific. Connection is dead.

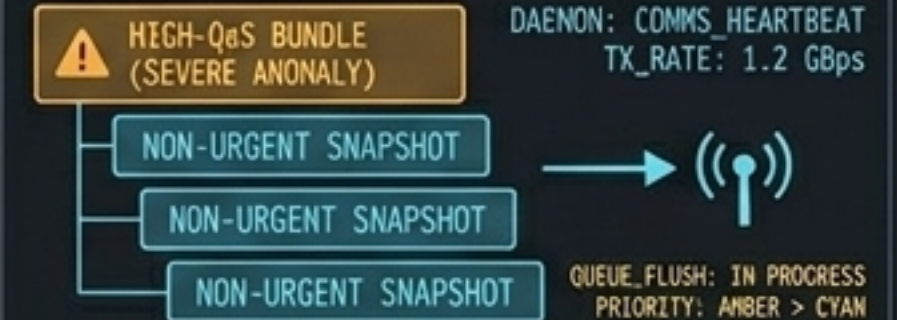
Step 2: Local SQLite Buffering



DTN (Disruption Tolerant Networking) logs alerts with sequence numbers, novelty scores, and severity prioritization.



Step 4: Priority Flush



High-QoS bundles (severe anomalies) are transmitted first via a Comms Daemon heartbeat, followed by non-urgent snapshots.

Step 3: Ground Station Handshake



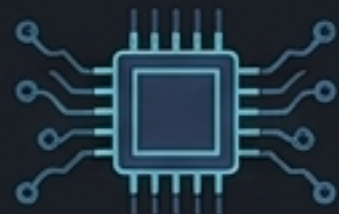
Satellite enters the Transmission Zone and establishes a secure link.

The Ground Segment: A zero-backend intelligence dossier



Offline-First PWA

Built on WebAssembly (wllama), the interface downloads a highly compressed text model to the local cache upon installation.



Absolute Data Privacy

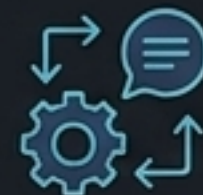
When the satellite downlinks a JSON coordinate alert, the operator queries the data locally. No cloud routing. No API fees. No third-party data leaks.



Agentic Tool Calling

[USER]: Cross-reference these coordinates with known facilities.

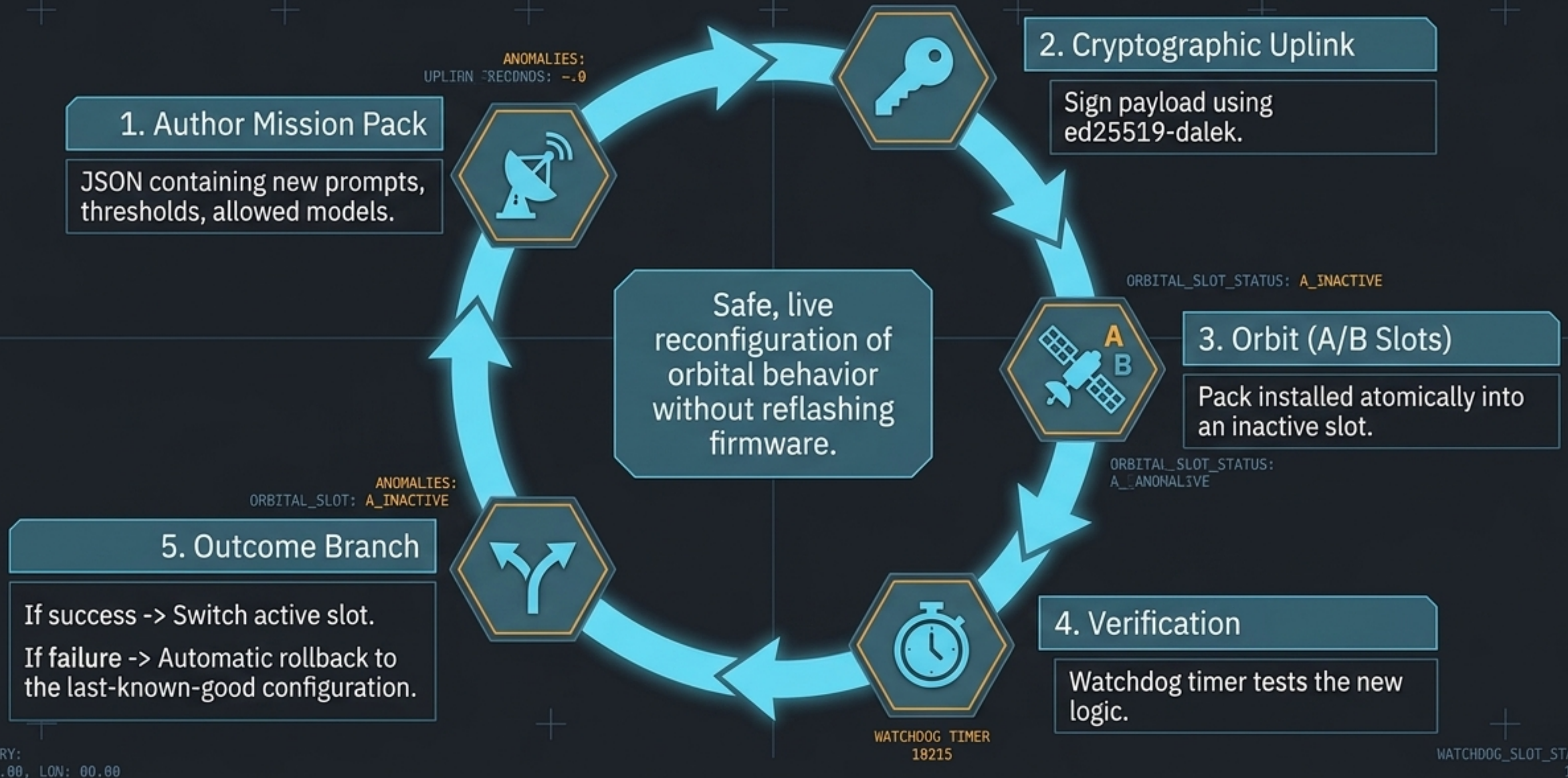
[SYSTEM]: Analyzing local basemaps. High probability of unmanaged wildfire.



Mission Packs: Live orbital reconfiguration without reflashing firmware

TELEMETRY: UPLINK_OK17

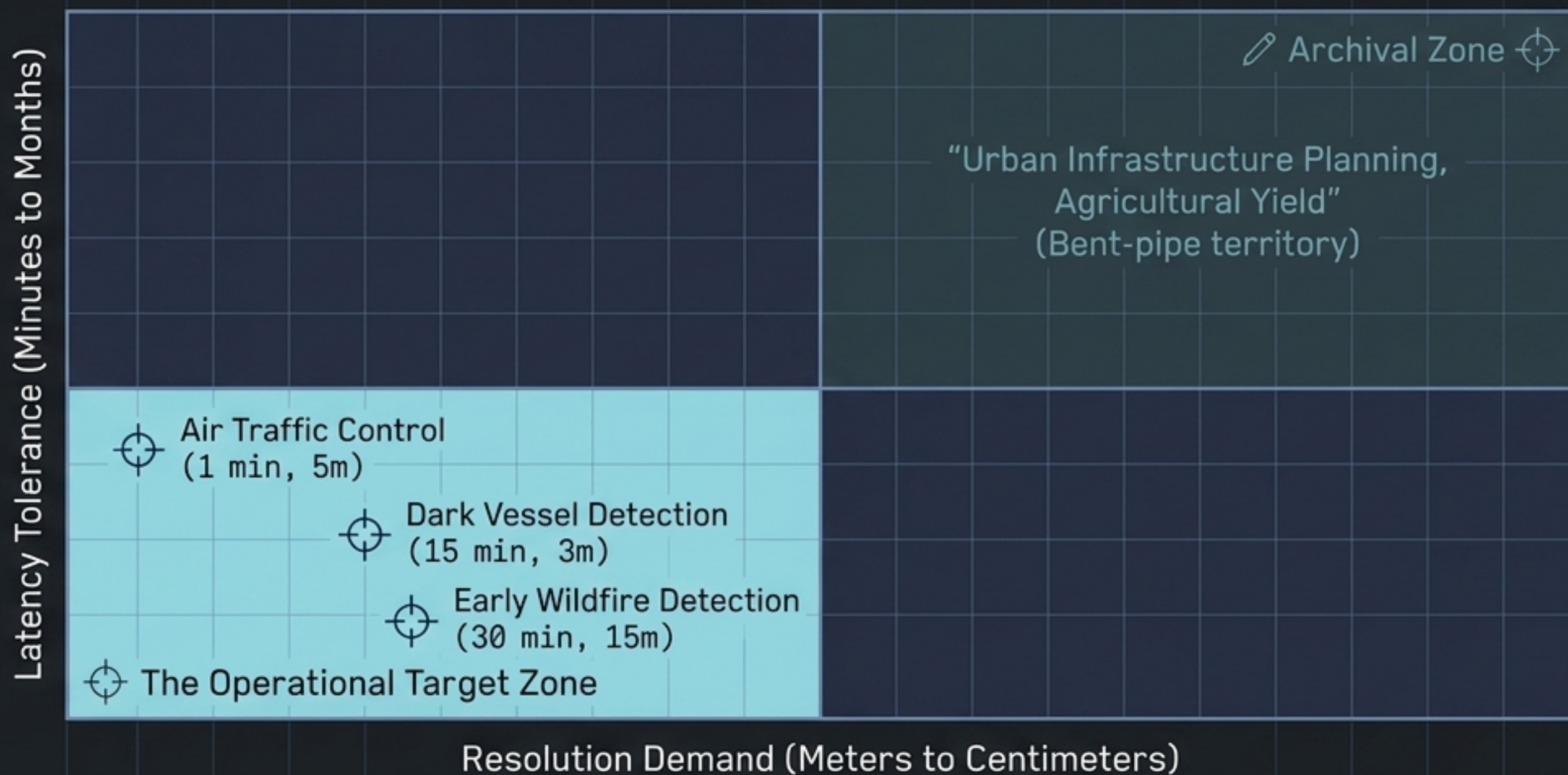
LAT: 88.00, LON: 00.00



TELEMETRY:
LAT: 88.00, LON: 00.00

WATCHDOG_SLOT_STATUS:
10215

Targeted for operational edge cases, not archival storage



Archival tasks tolerate latency for precision.
Tactical military and disaster response demand immediate alerting.

Operational Target I: Canopy Sentinel & Early Wildfire Detection



The Challenge

Coarse legacy sensors fail to detect small, nascent fires obscured by thick smoke and canopy cover in a timely manner.

The Orbital Solution

Ingests Sentinel-2 multispectral frames (thermal/infrared bands). The VLM performs multi-image comprehension across sequential frames to deduce spread direction and intensity.

The Result

Downlinks coordinate bounding boxes and natural-language risk assessments directly to state emergency dispatchers in under 30 minutes, circumventing traditional bent-pipe delays.

Operational Target II: Maritime Domain Awareness



The Challenge:

Illicit fishing and embargo smugglers disable Automatic Identification System (AIS) transponders to evade tracking.

The Orbital Solution:

- Utilizes onboard Synthetic Aperture Radar (SAR) combined with LFM2.5-VL to continuously scan the ocean surface, cataloging all physical marine vessels.
- It cross-references physical presence against a cached AIS registry.

The Result:

Instantly flags 'Dark Vessels' (Physical ship + No AIS signal) and downlinks exact coordinates and estimated headings to military Common Operating Picture (COP) systems.

057A
1785:
OETA
TFENDOOV
G0RL
SEABER
POVEEOSNO
SONET
—
TELENGTHY
SONEA

207A
2910F
110091
10103E
72.00
14644
0.000
301.1
9.260

00:00:018016001182719038

Operational Target II: Defense & Space-WATCH



The Challenge:

The DoD's DIU 'Ghost Recon' and DARPA 'Space-WATCH' programs require persistent tracking and autonomous characterization of objects in Low Earth Orbit (LEO).

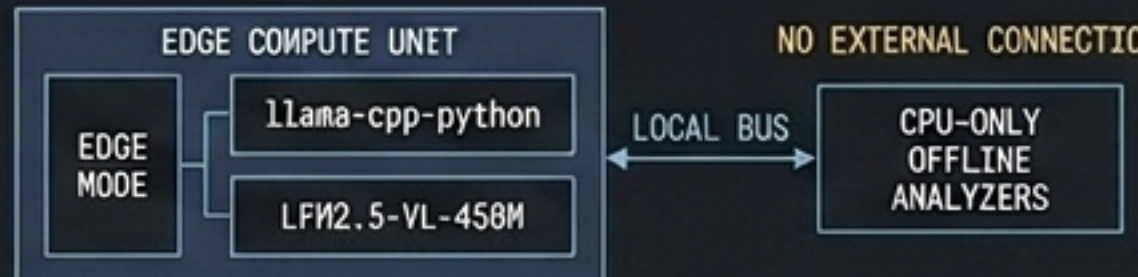
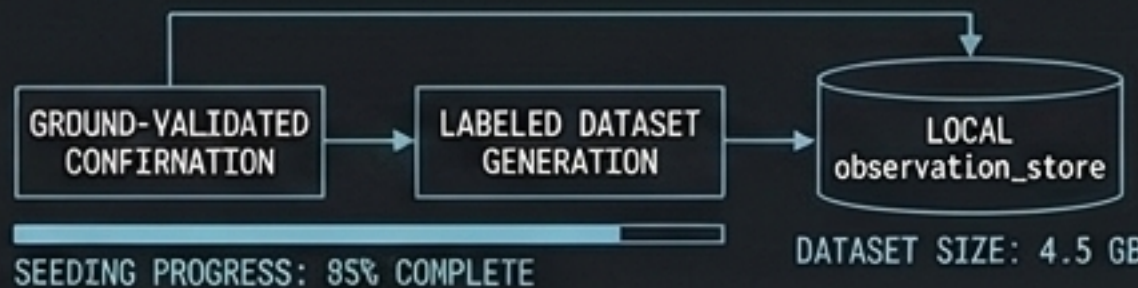
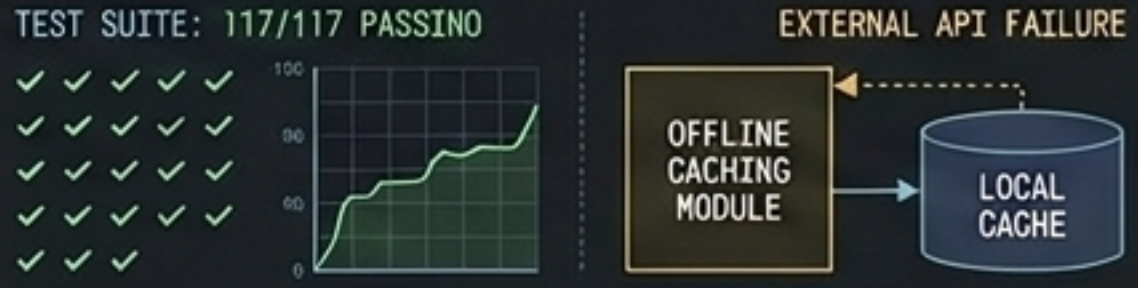
The Orbital Solution:

Edge-deployed LFM's provide autonomous space-to-space target characterization. Decentralized, open-weight models ensure that proprietary tracking algorithms (the model weights) remain strictly within the operator's control.

The Result:

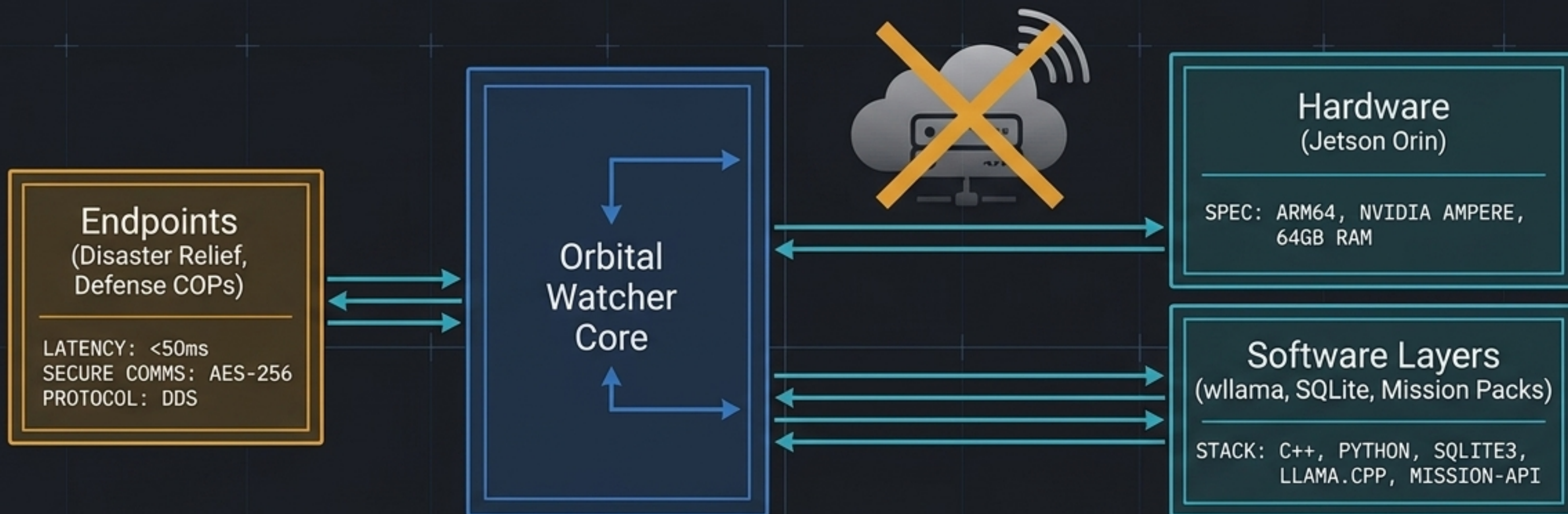
Continuous sub-synchronous monitoring and real-time anomalous maneuver alerting without dependency on foreign data analytics.

Proof of Operation: Fulfilling the Liquid Track criteria

ROW 1 [VERIFIED] ⌚ 09:25:45 UTC	Satellite Imagery Integration Seamlessly ingests DPhi SimSat API Sentinel-2 10m cloud-masked composites.	
ROW 2 [VERIFIED] ⌚ 09:25:45 UTC	Technical Implementation (Zero Cloud) Runs llama-cpp-python and LFM2.5-VL-450M locally. Ground analysis uses CPU-only offline signal analyzers. (Current implementation uses optional GGUF on dev hardware; full offline Jetson-class inference is the target.)	
ROW 3 [VERIFIED] ⌚ 09:25:45 UTC	Fine-Tuning Readiness Includes data seeding capability where ground-validated confirmations generate labeled dataset points saved to local observation_store for future leap-finetune optimization.	
ROW 4 [VERIFIED] ⌚ 09:25:45 UTC	Resiliency 117 Playwright E2E tests passing. Degrades gracefully to offline caching when external APIs fail.	

SYSTEM STATUS: NOMINAL
LATENCY: <30ms
TIMESTAMP: 2024.06.12 14:38:45 UTC

Synthesis: A closed-loop Operating System for Low Earth Orbit



Synthesis Panel

LFM-ORBIT is not simply an image classifier. It is a fundamental inversion of Earth Observation architecture.

It transforms the satellite from a passive recording instrument into an active, intelligent, autonomous sensor node—bridging the gap between raw optical data and actionable human intelligence without ever touching the cloud.

SYSTEM STATUS: NOMINAL
LATENCY: <20ms
TIMESTAMP: 2024.06.12 14:39:45 UTC

[illegible]

By bringing Liquid Foundation Models to the edge, LFM-ORBIT delivers the intelligence of a massive ground station inside a future 16 GB-class orbital compute node.